PRE-ALGEBRA (8TH) · UNIT 5: PATTERNS AND BIVARIATE DATA

Screen Time & Sleep Study

Use scatter plots, lines of fit, and two-way tables to study screen time and sleep.

CP1

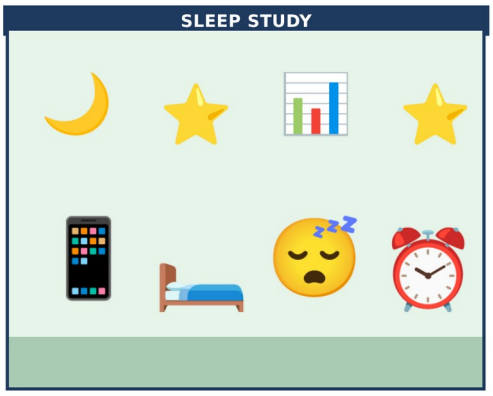
2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

The school health committee wonders whether screen time is connected to how much students sleep. They collected data from students and want a clear, honest report.

You will analyze the data with scatter plots, a line of best fit, and a two-way table, and present what the data can and cannot tell us.



Model

sleep ≈ −0.45 × screen + 9.4

(hours per night)

Survey

50 students

phone in bedroom? sleeps 8+ h?

Learning Goals

- I can describe patterns of association, clusters, and outliers.
- I can write and use a linear model and compute residuals.
- I can use relative frequencies to describe association in a two-way table.

Project Timeline

When	What you will do
Day 1	Patterns
Day 2	Predictions & Residuals
Day 3	Relative Frequencies
Final	Research Poster + reflection

Part 1: Patterns

Day 1 · Use the Study Data card.

Study Data

Screen time (h) → Sleep (h)
1 → 9.0 · 2 → 8.6 · 2.5 → 8.4
3 → 8.0 · 4 → 7.6 · 4.5 → 7.5
5 → 7.0 · 6 → 6.6 · 6.5 → 6.4 · 8 → 5.8

Two-way table (50 students)
Phone in bedroom: 6 sleep 8+ h, 24 sleep < 8 h
No phone in bedroom: 14 sleep 8+ h, 6 sleep < 8 h

1. Describe the direction, form, and strength of the association.

2. Draw a line through (1, 9) and (8, 5.8). Find its slope and equation.

3. Compare your line with the model y = −0.45x + 9.4. Are they close? Explain.

Part 2: Predictions & Residuals

Day 2 · Use y = −0.45x + 9.4.

4. Predict sleep for 5.5 hours of screen time.

5. Find the residual for the student with 5 hours of screen time and 7.0 hours of sleep.

6. Use the model to predict sleep for 20 hours of screen time. Is this prediction reasonable?

Part 3: Relative Frequencies

Day 3 · Use the survey of 50 students.

7. Make a row relative-frequency table (percents within each row).

8. What percent of all 50 students have a phone in the bedroom and sleep 8+ hours?

9. Does the table prove that phones in bedrooms cause less sleep? Explain.

Final Product: Research Poster

Your final product must include:

- Scatter plot with a hand-drawn line and the model
- Predictions and a residual
- Relative-frequency table
- Discussion of association vs. causation
- Research poster for the health committee

Reflection

R1. What does a negative residual mean?

R2. Why is extrapolation risky?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Scatter plot, model, predictions, and two-way analysis are accurate and correctly interpreted. Goes beyond the minimum with a creative, realistic choice.	Scatter plot, model, predictions, and two-way analysis are accurate and correctly interpreted.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Poster states findings clearly and explains limits (association vs. causation, extrapolation). Polished, organized, and visually engaging.	Poster states findings clearly and explains limits (association vs. causation, extrapolation). Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.